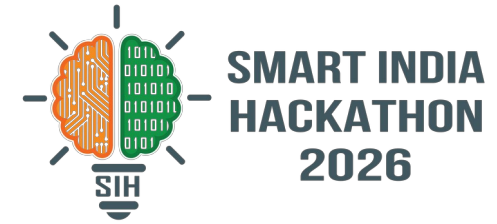


SMART INDIA HACKATHON 2026



RASTA AI

Route Intelligence for Essential-Supply Logistics in North Eastern India

SIH26002

Problem Statement ID

MDoNER • Smart Automation • Software

PROBLEM STATEMENT ID

SIH26002

PROBLEM STATEMENT TITLE

AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)

THEME

Smart Automation

PS CATEGORY

Software

TEAM ID

17 (internal group no.)

TEAM NAME

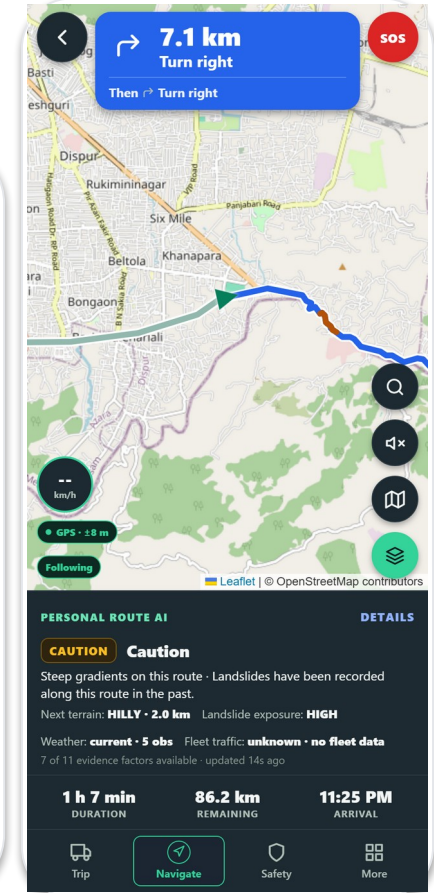
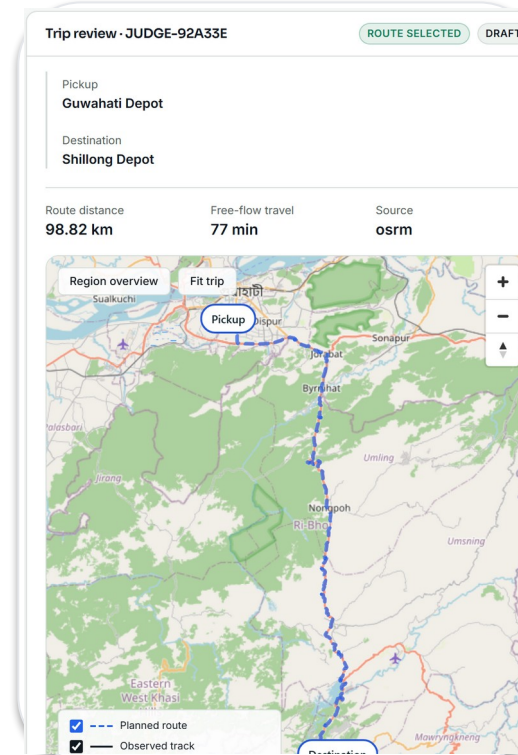
NER-AI LOGISTICS

SHORTEST IS NOT ALWAYS USABLE.

“Can this essential-supply truck reliably use this corridor now?”

End-to-end working prototype • physical-phone validated • live demo corridor Guwahati → Shillong

LIVE WORKING SYSTEM manager console + Android driver app



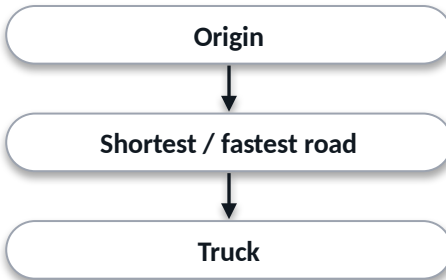
Manager route review • Driver navigation — real screens, real road

PROPOSED SOLUTION

NOT THE SHORTEST ROAD. THE ROAD THAT IS OPERATIONALLY USABLE NOW.

NER corridors: mountain terrain • monsoon • landslides • floods • weak signal • stale evidence. **A shortest-path router still sends the truck.**

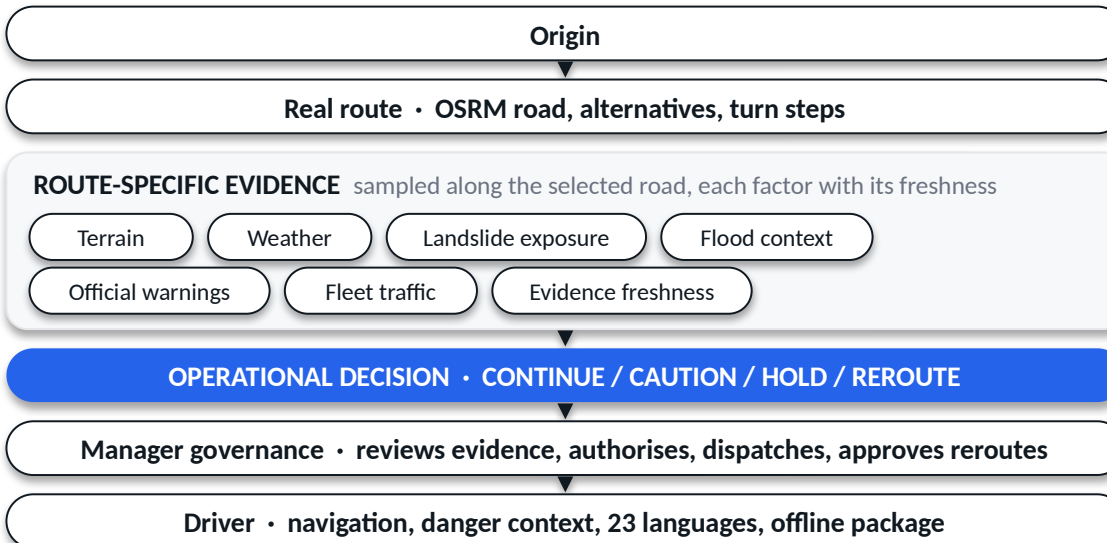
A NORMAL ROUTER



No terrain, weather, hazard or freshness check.

No governance. No answer when data is missing.

RASTA AI evidence before the road



UNKNOWN ≠ SAFE

The system refuses to fabricate certainty: missing or stale evidence is marked UNKNOWN, never counted as safe, and the route goes to an authorised reviewer.

MANAGER • CHECK CONDITIONS

REQUIRES REVIEW

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Landslide data is not available - no source is connected. • Steep gradients on this route • Landslides have been recorded along this route in the past. • The landslide inventory ends in 2017 - recent years are not covered. • River levels near this route are close to their recent average. • No official disaster alert names a district on this route right now. • No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Road quality, Truck restrictions

Evidence coverage • Weather AVAILABLE • Terrain AVAILABLE • Warnings AVAILABLE • Traffic UNKNOWN

TERRAIN **0.5 KM STEEP**

49–1436 m • 1975 m climb • steepest 10.3%
flat 61.8 km • rolling 29.6 km • hilly 7.0 km • steep 0.5 km
Copernicus DEM GLO-90 via Open-Meteo • SRTM 30 m via OpenTopoData • 100% of 198 samples answered

LANDSLIDE HISTORY **HIGH EXPOSURE**

22 recorded landslides within 5 km of the road, nearest 0.4 km
6 more nearby placed too imprecisely to count • HIGH is three or more; MODERATE is one or two
NASA Global Landslide Catalog • inventory 2007–2017 — aged, recent years not covered

RIVER LEVELS

NORMAL

Discharge at 5 river cells along the corridor, highest 1.0× its own 30-day mean • GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

OFFICIAL ALERTS

NONE ON CORRIDOR

NDMA SACHET CAP feed, 99 alerts nationwide at 10:13:54 PM • placed by district name (Dispur, East Khasi Hills, Kamrup Metropolitan, Myllem, Ri-Bhoi, Umiling, Umsning) • 4 more active elsewhere in the corridor states, not placeable on this road.

FLEET TRAFFIC

UNKNOWN

Real screen • evidence • freshness • UNKNOWN

TECHNICAL APPROACH

VERIFIED DATA → DECISION → DRIVER provider adapters with health + freshness

WEATHER

Open-Meteo · MET
Norway

TERRAIN

Copernicus DEM ·
OpenTopoData

WARNINGS

NDMA SACHET (CAP)

FLOOD

GloFAS

**LANDSLIDE
HISTORY**

NASA GLC 2007-17

TRAFFIC

fleet GPS fixes

ROUTE-SPECIFIC EVIDENCE · FastAPI backend

every factor sampled along the selected road · PostgreSQL + PostGIS (Supabase) · OSRM routes, alternatives, turn steps · freshness per factor

DETERMINISTIC SAFETY POLICY

11 factors · reason codes · UNKNOWN ≠ SAFE · reviewer authorisation when hazard data is missing

CONTINUE**CAUTION****HOLD****REROUTE****MANAGER REVIEW****DRIVER NAVIGATION****ROUTE-AHEAD MONITOR (60 s)****CONDITIONS CHANGE?****REASSESS / REROUTE**

Decision loop: conditions are re-scored every 60 s along the road ahead; a material change goes back through the policy, and a reroute is proposed to the manager — never applied silently.

BUILT WITH

CLIENTS Manager console — React + TypeScript, manager accounts only · Driver app — React Native / Expo, Android; one login, server-decided role, offline trip package

STACK FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini

EVIDENCE Open-Meteo / MET Norway · NDMA SACHET · GloFAS · NASA historical landslides · OpenTopoData / Copernicus DEM

WHO DECIDES AI is supporting

DETERMINISTIC SAFETY ENGINE

makes every operational decision

GEMINI / OPENROUTER (LLM)

explains the decision and assists in the driver's language — never decides

EXPERIMENTAL ML

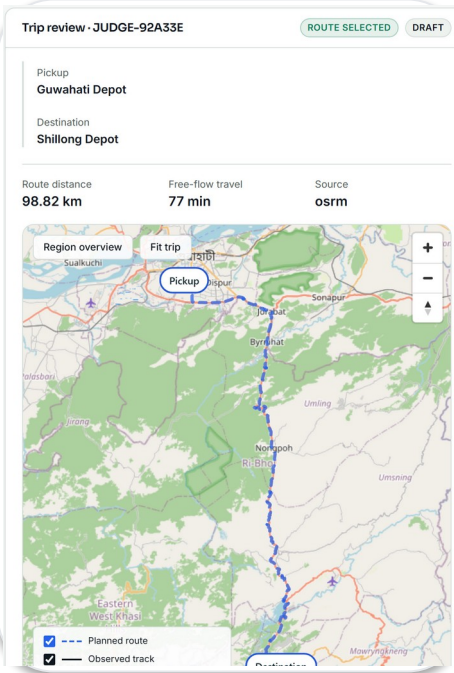
research channel only; kept outside routing by the safety-validation gate

GOVERNANCE CHAIN**Verified evidence****Deterministic safety policy****Manager review****Driver action**

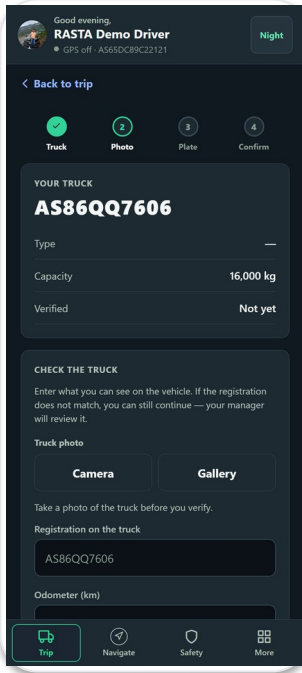
NOT A CONCEPT. A WORKING END-TO-END PROTOTYPE.

PLAN → REVIEW → DISPATCH → ACCEPT → TRUCK VERIFY → NAVIGATE → OFF-ROUTE → MANAGER REROUTE APPROVAL → DELIVERY

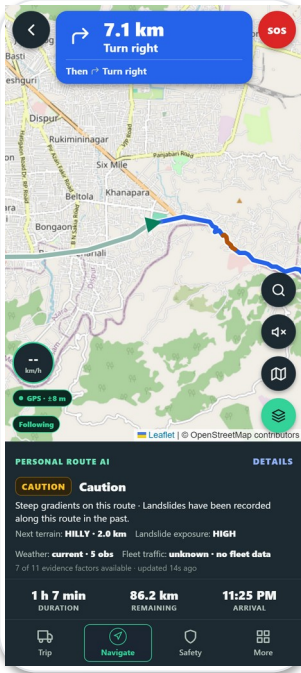
REAL SCREENS FROM THE CERTIFIED RUN



Manager • route review



Driver • truck check



Driver • navigation

PROOF physical Android phone • hosted backend • real road

The whole lifecycle above was run end to end on a physical Android phone against the hosted backend, on the real Guwahati → Shillong road.

12/12 physical judge-flow steps

9/9 physical role-flow steps

1138+ backend • **621+** driver • **170+** manager tests

PHYSICAL ANDROID • CERTIFIED

Canonical demo Guwahati → Shillong • ≈ 98.8 km real route

Every reroute on that run was approved by a manager before the phone followed it.

KNOWN LIMITS → MITIGATION Render cold start → warm /health before the demo • provider rate limits → cached evidence + explicit UNKNOWN • background push needs Firebase config → in-app alerts already work • experimental ML → outside routing until the gate passes

IMPACT AND BENEFITS

RASTA AI does not ask only “Which road is shortest?” It asks “**Can this truck reliably use this corridor now?**”

FOOD • MEDICINE • FUEL • RELIEF SUPPLIES — essential logistics for hill communities, decided on evidence

WHO BENEFITS

FLEET MANAGERS

Route usability and uncertainty visible before dispatch; auditable decisions.

DRIVERS

Safer navigation, danger context, emergency support, 23 languages.

REMOTE COMMUNITIES

More resilient access to food, medicine, fuel and relief supplies.

GOVERNMENT / OPERATIONS

Auditable route evidence and disruption visibility per corridor.

DEMONSTRATED TODAY

23 selectable languages, explicit fallback status

11 route factors in every decision

60 s route-ahead monitor while driving

12/12 judge-flow steps certified on a physical phone

100 % of reroutes reviewed by a human

0 unknown factors counted as safe

HUMAN GOVERNANCE AI has no uncontrolled authority • no invented percentages

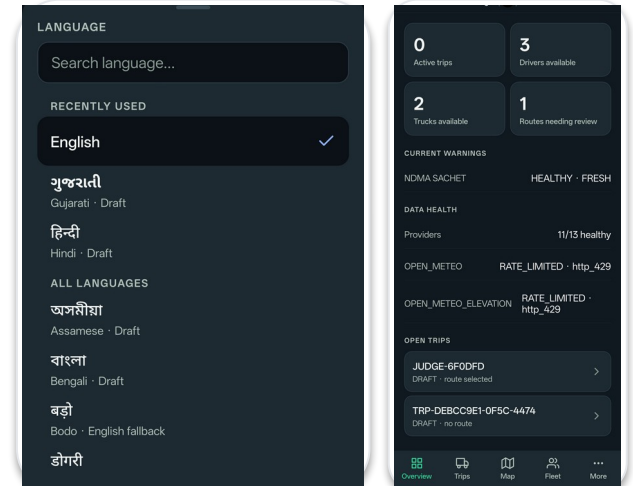
Verified evidence

Deterministic safety policy

Manager governance

Driver action

REAL SCREENS • APK 1.0.18



Driver • 23 languages

Manager • mobile

Real screens, APK 1.0.18. One login, server-decided role: a manager gets a mobile fleet view, a driver gets navigation. Each language shows its status — Verified, Draft or English fallback — so nobody is misled.

RESEARCH / EVIDENCE SOURCES BEHIND THE ROUTE DECISION

ROUTING / PLACES

OSRM • OpenStreetMap • Nominatim

real road geometry, alternatives and turn steps; place search on the corridor

OFFICIAL / FLOOD

NDMA SACHET (CAP) • Copernicus GloFAS

official warnings polled with freshness; river discharge against its 30-day mean

AI ASSISTANCE

Google Gemini

explains the decision in the driver's language — never safety authority

WEATHER / TERRAIN

Open-Meteo • MET Norway • Copernicus DEM • OpenTopoData

forecast sampled along the road; climb, gradient and steep segments (GLO-90)

LANDSLIDE HISTORY

NASA Global Landslide Cataloghistorical exposure; inventory 2007–2017, ≤ 5 km location-accuracy filter

PROBLEM STATEMENT

MDoNER • SIH26002

AI-based smart logistics and accessibility intelligence for the NER

EXPERIMENTAL LANDSLIDE RESEARCH

95 %**RECALL**

on the NER geographic holdout — 154 catalogued events, no NER row used in training

SAFETY GATE

NOT DEPLOYED**FPR 50 %** too high for production deployment

TRAIN

HELD-OUT NER TEST

USEFUL SIGNAL FOUND

FALSE POSITIVES TOO HIGH

SAFETY GATE

NOT DEPLOYED

VALIDATION BEFORE AUTOMATION. Strong recall was not enough. Held-out testing exposed the false-positive rate, so the gate keeps the model out of routing.

Precision 0.32 • F1 0.48 • PR-AUC 0.55 • ROC-AUC 0.83 • Brier 0.12 • calibrated logistic regression on NASA GLC + ERA5-Land rain + DEM slope • docs/HAZARD_ERROR_ANALYSIS.md

REFERENCES

gpm.nasa.gov/landslides (NASA GLC) • sachet.ndma.gov.in • global-flood.emergency.copernicus.eu (GloFAS) • open-meteo.com • api.met.no • opentopodata.org • Copernicus DEM • project-osrm.org • openstreetmap.org • github.com/nxtlucifer/ner-ai-logistics (repository, tests, model registry, error analysis)